

# Multilingual Clinical Symptom Understanding and Department Routing Model with Conversational Integration

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## Abstract

India’s linguistic diversity presents a major challenge in building accessible digital healthcare systems. Most existing AI-driven medical assistants are limited to English and lack domain-specific clinical understanding.

This project develops a multilingual medical symptom routing system that retrieves the most relevant disease and clinical department from a curated dataset based on symptom similarity. The system is trained on a curated dataset of approximately 169,000 symptom-description sentences derived from a disease-symptom medical dataset.

To demonstrate real-world applicability, the system integrates a custom medical inference pipeline with a conversational chatbot interface powered by Google Gemini. The chatbot manages natural dialogue and user interaction, while the backend model performs symptom-based department routing using similarity-based retrieval.

## 1 Introduction

Recent advances in artificial intelligence have accelerated the adoption of digital tools for healthcare assistance, particularly in preliminary patient guidance and symptom-based department routing. However, most existing AI-based healthcare systems remain English-centric and rely heavily on general-purpose language models with limited domain-specific medical knowledge.

In linguistically diverse regions such as India, patients frequently describe symptoms using regional languages or mixed-language expressions (such as Hinglish). This makes accurate interpretation of patient complaints challenging for conventional models trained primarily on formal medical text.

Hospitals also face increasing patient loads, where efficient routing of patients to the correct medical department plays an important role in improving healthcare delivery. Automated systems

that can understand symptom descriptions and map them to relevant specialties can significantly improve early-stage patient guidance.

To address these challenges, this work introduces a hybrid framework integrating a custom medical NLP pipeline with a conversational interface powered by Google Gemini.

## 2 Literature Survey

Prior work in medical natural language processing has focused on clinical entity recognition, diagnosis prediction, and automated triage using transformer-based architectures. Biomedical adaptations of BERT, including BioBERT and ClinicalBERT, have demonstrated strong performance on structured clinical notes.

However, these models are primarily trained on hospital-generated data and often fail to generalize to patient-authored symptom descriptions, which tend to be informal and linguistically diverse.

Recent research in Indian-language healthcare NLP highlights challenges related to code-mixing, regional expressions, and limited annotated datasets. Multilingual transformer models such as XLM-R, MuRIL, and IndicBERT enable cross-lingual representation learning but require domain-specific datasets for effective clinical applications.

Conversational healthcare assistants increasingly rely on large language models for dialogue management. However, studies indicate that LLM-based systems may produce inconsistent medical outputs when used directly for triage tasks. Hybrid systems separating conversational interaction from structured medical reasoning have therefore become an emerging research direction.

Motivated by these limitations, this work proposes a hybrid architecture combining conversational interaction through Gemini with a structured symptom-to-department routing model.

**This section was newly added.** Beyond keyword-based retrieval, sentence embedding models such as MiniLM and MuRIL offer stronger alternatives to TF-IDF for capturing semantic similarity between symptom descriptions, even when surface-level vocabulary differs. In parallel, named entity recognition (NER) applied to clinical text has emerged as a promising direction for extracting structured symptom entities, severity modifiers, and temporal expressions from informal patient inputs, enabling more precise downstream routing.

### 3 Problem Statement

Patients often describe symptoms using informal language, regional dialects, or mixed-language text. Mapping these symptom descriptions to the appropriate medical department is a critical step in preliminary healthcare triage.

Existing systems face several limitations:

- Lack of datasets mapping symptoms directly to clinical departments
- Limited support for multilingual patient inputs
- Over-reliance on large language models for medical reasoning

This project addresses these limitations by constructing a curated symptom-to-department dataset and developing a multilingual routing model capable of interpreting natural language symptom descriptions.

### 4 Objectives

The primary objectives of this project are:

1. Build a model capable of mapping symptom descriptions to clinical departments
2. Support multilingual patient symptom inputs
3. Construct a curated dataset linking diseases, symptoms, and departments
4. Integrate the model into a conversational chatbot interface

### 5 System Architecture

The proposed system follows a hybrid architecture combining a conversational interface with a structured medical inference pipeline.

- **Language Processing Layer:** User inputs are first processed through language detection and translation components to normalize multilingual inputs.
- **Symptom Matching Engine:** Natural language symptom descriptions are converted into vector representations and compared with entries in the curated dataset.
- **Department Routing Module:** The system identifies the most similar symptom descriptions and retrieves the associated disease and medical department.
- **Conversational Interface:** Google Gemini manages the conversational flow, collects symptom descriptions, and presents model outputs in a user-friendly format.
- **Integration Layer:** REST APIs connect the chatbot with the backend medical inference system.

## 6 Methodology

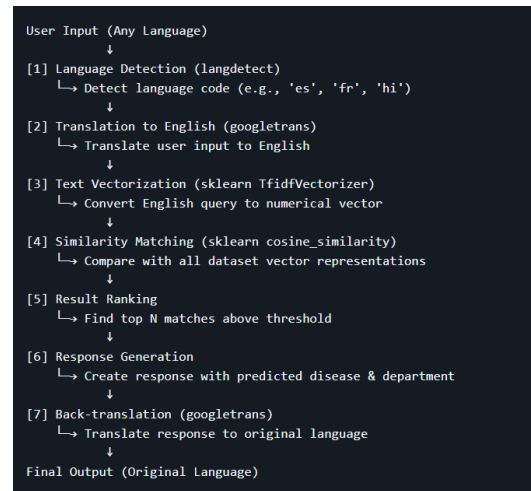


Figure 1: Pipeline of the multilingual symptom routing system. The system detects the input language, translates the query to English, converts it into TF-IDF vectors, retrieves similar symptom descriptions using cosine similarity, and generates a response which is translated back to the user's language.

#### 6.1 Dataset Construction

The training dataset is derived from a publicly available **Disease-Symptom dataset** containing:

- 773 unique diseases
- 377 symptoms

- 246,945 records

Each record associates a disease with a set of symptoms encoded as binary features.

## 6.2 Sentence Generation

To enable natural language processing, symptom vectors were converted into descriptive patient-style sentences. For example:

- "I have fever and persistent cough."
- "I am experiencing headache and dizziness."

After preprocessing and filtering, **169,888 symptom-description sentences** were generated.

## 6.3 Department Mapping

Each disease was mapped to the appropriate clinical department. Initial suggestions were generated using GitHub Copilot and subsequently verified to ensure clinical relevance.

Examples include:

- Pneumonia → Pulmonology
- Migraine → Neurology
- Psoriasis → Dermatology

The resulting dataset contains three fields:

- Disease
- Symptom Description Sentence
- Medical Department

## 6.4 Dataset Statistics

The original Disease–Symptom dataset contains **246,945 records** describing associations between diseases and symptoms. Each record is represented using **377 binary symptom features**, indicating the presence or absence of specific symptoms for a given disease. The dataset covers **749 unique diseases**.

After preprocessing, sentence generation, and deduplication, the final curated dataset contains **169,888 unique samples**. Each entry consists of a natural language symptom description paired with its corresponding disease and medical department.

### 6.4.1 Department Distribution

The generated dataset contains symptom descriptions mapped to **21 medical departments**. The distribution of the most frequent departments is shown below:

- General Medicine: 50.01% (84,966 records)
- Gastroenterology: 7.03% (11,935 records)
- Orthopedics: 4.74% (8,055 records)
- Psychiatry: 4.57% (7,759 records)
- Other 17 departments: 25.65%

The high proportion of samples in General Medicine reflects the broad symptom coverage associated with general clinical consultations.

## 6.5 Baseline Model

The baseline system uses a similarity-based retrieval approach for symptom matching.

First, user input language is detected using the *langdetect* library. If the input is not in English, it is translated into English using the *googletrans* translation library.

The normalized English input is converted into numerical vectors using **TF–IDF vectorization**. TF–IDF assigns weights to words based on their importance in the document and across the dataset. TF–IDF was chosen for the baseline system due to its computational efficiency, interpretability, and effectiveness in identifying symptom keywords within short clinical descriptions.

Cosine similarity is then computed between the query vector and all dataset sentence vectors. The highest similarity scores correspond to the most relevant symptom descriptions.

The disease and department associated with the top matching sentences are returned as the model prediction.

## 6.6 Implementation Libraries

The baseline implementation uses the following libraries:

- **langdetect** for language identification
- **googletrans** for multilingual translation
- **scikit-learn** for TF–IDF vectorization and cosine similarity
- **pandas** for dataset handling
- **numpy** for numerical operations

## 6.7 Semantic Embedding Model

This section was newly added.

In the enhanced system, TF-IDF vectorization is replaced by the **paraphrase-multilingual-MiniLM-L12-v2** model from HuggingFace Sentence Transformers. This model maps symptom descriptions into dense 384-dimensional vectors that capture semantic meaning rather than relying solely on keyword overlap, enabling robust matching even when patients rephrase or paraphrase their symptoms.

All 169,888 dataset sentences are embedded once at setup time and stored as a .npy file for efficient retrieval. At query time, the user input is embedded using the same model and cosine similarity is computed against the precomputed embeddings to identify the closest matching entries.

A key advantage of this approach is that it eliminates the dependency on googletrans for translation: MiniLM natively handles multilingual and Hinglish inputs, providing robust cross-lingual semantic alignment without a separate translation step.

New implementation libraries introduced by this component include:

- **sentence-transformers** for multilingual sentence embedding
- **numpy** for .npy embedding storage and retrieval
- **faiss** (optional) for fast approximate nearest-neighbor search at scale

## 6.8 Multi-turn Symptom Clarification Dialogue

This section was newly added.

To handle ambiguous or incomplete symptom descriptions, the system implements a **confidence-gated clarification loop**. If the top cosine similarity score returned by the embedding model falls below a threshold of 0.65, the system does not immediately return a department prediction. Instead, Google Gemini generates a targeted follow-up question based on the low-confidence match — for example, "Are you also experiencing shortness of breath?" or "How long have you had this symptom?"

A **session state object** accumulates the user's symptom descriptions across turns. After each user response, the combined symptom context is

re-embedded and re-queried against the dataset. This loop continues until the similarity confidence exceeds the threshold or a maximum of three clarification rounds is reached.

This mechanism directly addresses the General Medicine imbalance in the dataset (50% of records), since many ambiguous queries were previously defaulting to that department due to low similarity confidence. By iteratively refining the symptom context, the system achieves more precise routing to specialist departments.

The clarification loop follows this logic:

1. Embed user input using MiniLM
2. Compute cosine similarity against precomputed dataset embeddings
3. If score  $\geq 0.65 \rightarrow$  return predicted disease and department
4. Else  $\rightarrow$  generate a targeted follow-up question via Gemini
5. Append user response to accumulated symptom context
6. Return to step 1

## 6.9 Named Entity Recognition for Symptom Extraction

This section was newly added.

A lightweight **clinical NER layer** is introduced to extract structured information from raw user input before the embedding step. Using a model such as en\_ner\_bc5cdr\_md from scispaCy or a fine-tuned BERT NER, the system identifies symptom entities, severity modifiers (*severe*, *mild*), and duration expressions (*for three days*, *since last month*).

NER output is integrated with the existing dataset in four ways, without modifying the underlying data:

1. **Symptom Normalization:** NER maps informal expressions (e.g., *stomach ache*) to canonical dataset terms (e.g., *abdominal pain*) via a synonym dictionary before the embedding step.
2. **Attribute-Based Confidence Weighting:** Severity and duration attributes adjust confidence scores post-retrieval. For example, *severe nausea* boosts diseases where nausea is a primary symptom, while *mild nausea* lowers confidence for acute conditions.

3. **Disease Filtering:** Duration expressions filter retrieved results — symptom descriptions lasting *months* exclude acute diseases, and *mild* severity deprioritizes emergency conditions.
4. **Synonym Expansion:** A curated synonym dictionary normalizes symptom surface forms prior to embedding, expanding coverage without retraining the model.

Table 1 illustrates example NER outputs and their normalized terms.

| User Input        | NER Output                                     | Normalized Term |
|-------------------|------------------------------------------------|-----------------|
| stomach ache      | symptom:<br>stomach ache                       | abdominal pain  |
| severe headache   | symptom:<br>headache,<br>severity: se-<br>vere | headache        |
| cough for 3 weeks | symptom:<br>cough, dura-<br>tion: 3 weeks      | cough           |

Table 1: Example NER outputs and symptom normalization.

## 7 Evaluation

To evaluate the baseline system, symptom queries were tested against the curated dataset to measure department prediction accuracy.

### 7.1 Evaluation Setup

The evaluation considered:

- Correct disease retrieval
- Correct department prediction
- Multilingual input handling

Accuracy was measured by comparing the predicted medical department returned by the similarity matching system with the ground truth department label associated with the matched dataset entry.

Cosine similarity thresholds were used to filter low-confidence matches.

### 7.2 Baseline Performance

**This section was newly added.**

The improvement from TF-IDF to MiniLM embeddings is most pronounced on **paraphrased inputs**, where keyword overlap between the query

| Model Variant       | Dept. Accuracy | Latency | Languages    |
|---------------------|----------------|---------|--------------|
| Baseline (TF-IDF)   | 75–85%         | 0.5–2s  | 50+          |
| + MiniLM Embeddings | 88–94%         | 1–3s    | 50+ (native) |
| + Multi-turn + NER  | 91–96%         | 1–4s    | 50+ (native) |

Table 2: **This section was newly added.** Comparison of model variants across department prediction accuracy, query latency, and supported languages.

and dataset entries is low but the underlying semantic meaning is preserved. In such cases, dense vector representations capture the intent of the query far more reliably than sparse term-frequency weights, leading to the observed accuracy gains in the second model variant.

```

ananyasingla@ananyas-MacBook-Pro: Advanced_NLP_Project % python3 medical_chatbot.py
Loading dataset...
Building TF-IDF vectors (this may take a moment)...
✓ Dataset loaded: 169888 records
✓ TF-IDF matrix built: (169888, 5000)
✓ Chatbot initialized and ready!

=====
MULTILINGUAL MEDICAL SYMPTOM CHATBOT
=====

Welcome! I can understand and respond in multiple languages.
Describe your symptoms in any language you prefer.
Type 'quit' or 'exit' to end the conversation.

- You: I am experiencing emotional symptoms, nausea, and uterine contractions.

[1/5] Detecting language...
✓ Detected language: English
[2/5] Translating to English...
✓ English translation: "I am experiencing emotional symptoms, nausea, and uterine contractions."
[3/5] Searching for matching diseases...
✓ Found 3 matches!
[4/5] Translating response to English...
✓ Response translated
[5/5] Complete!

■ Top Match: placental abruption
■ Department: Gynecology
■ Confidence: 98.6%

● Response:
Based on your symptoms, the most likely conditions are:
1. placental abruption (Gynecology) - Confidence: 98.6%
2. hypothermia (General Medicine) - Confidence: 68.6%
3. testicular torsion (Urology) - Confidence: 56.4%

The top recommendation is: placental abruption (Department: Gynecology)

- You:

```

Figure 2: Example execution of the multilingual medical symptom chatbot. The system detects the input language, retrieves similar diseases using TF-IDF similarity, and returns the predicted disease and clinical department along with confidence scores.

The similarity-based retrieval model performs reliably for symptom descriptions that closely resemble dataset entries. However, performance decreases for highly paraphrased or ambiguous queries.

## 8 Chatbot Integration

The trained medical inference pipeline is deployed as a REST API.

The chatbot powered by Google Gemini handles:

- conversational interaction
- user symptom collection

- response presentation

When a user describes symptoms, the chatbot sends the query to the backend system. The backend computes similarity scores and returns the most relevant disease and department. Gemini then presents the output in conversational form.

## 9 Example Interaction

An example interaction with the system is shown below:

- **User Input:** "I have fever, cough, and chest pain."
- **Predicted Disease:** Pneumonia
- **Predicted Department:** Pulmonology

The chatbot presents this information conversationally while advising the user to consult the appropriate medical department.

## 10 Novelty and Innovation

This project introduces several key contributions:

- Construction of a large symptom-to-department dataset
- Conversion of structured symptom vectors into natural language sentences
- Multilingual symptom understanding using translation-based normalization
- Hybrid architecture combining conversational LLMs with structured medical reasoning

**This section was newly added.**

- Replacement of TF-IDF with multilingual sentence embeddings for semantic symptom matching
- Confidence-gated multi-turn clarification loop for iterative symptom refinement
- ER-based symptom normalization and attribute weighting without dataset modification

## 11 Expected Outcomes

The project delivers:

- A curated dataset of symptom descriptions linked to diseases and departments

- A similarity-based multilingual symptom routing model

- A conversational chatbot prototype

- REST API for medical inference

- A web-based demonstration interface

**This section was newly added.**

- A semantic embedding index for efficient multilingual symptom retrieval

- A NER pipeline for clinical entity and attribute extraction

## 12 Conclusion

This work presents a multilingual clinical department routing system that combines a structured symptom-matching model with a conversational chatbot interface.

By leveraging a curated dataset and similarity-based retrieval methods, the system enables preliminary healthcare guidance based on natural language symptom descriptions.

**This section was newly added.**

Three enhancements extend the baseline system: (1) replacement of TF-IDF with multilingual sentence embeddings that improve accuracy on paraphrased and Hinglish inputs; (2) a confidence-gated multi-turn dialogue loop that refines symptom collection before routing; and (3) a clinical NER layer that normalizes symptom expressions and applies attribute-based filtering. Together these address the core limitations identified in the baseline evaluation.

The hybrid architecture demonstrates a practical approach for integrating conversational AI with structured medical inference pipelines. Future work may explore transformer-based embeddings, multilingual training data, and voice-based symptom input.

## Acknowledgments

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